

Final Report

Smart Phone Data analysis using R

Fundamentals Of Data Analytics

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Introduction:

The dataset chosen has a collection of data of smartphone models. This data is extracted from SmartPrikes sites like Selenium and Beautiful Soup. The dataset, "smartphone-uncleaned.csv," contains information about various features of smartphones including model, customer’s rating, technical specifications, prices, battery and memory of the smart phone. The dataset contains 1020 model of smartphones from various brands. This report involves the patterns that we analysed by using r. The report includes the patterns and trends in the pricing and the features of the mobile phone, relationship between user ratings and specifications in the smartphone, and understanding the preference of the customers in features and price range.

Data Overview

The dataset contains 1020 entries and 11 columns:

- **Model:** Smartphone model name.
- **SIM:** SIM support and network compatibility.
- **Price:** Price of the smartphone (string format with currency symbols).
- **Display:** Display size, resolution, and refresh rate.
- **Rating:** User rating out of 100.
- **OS:** Operating system version used in the smart phone.
- **Processor:** Details about the processor (brand, type, speed).
- **RAM:** RAM and internal storage capacity (string with units).
- **Battery:** Battery capacity and fast charging support (string format).
- **Camera:** Camera specifications (rear and front cameras).
- **Card:** Memory card support and capacity.

Data Cleaning Process:

For the analysis of the dataset, cleaning the dataset should be done. For the cleaning of data, the following steps have been used:

- **Price conversion:** We removed currency symbols and converted prices into numeric format. i.e., from “â,154,999” to “54,999”
- **RAM and storage separation:** Split the RAM and storage into separate columns for easier analysis. This division of data can also be used for analysing and getting graphical outputs like bar graph.

Before Cleaning the data:

	ram
	12â€GB RAM, 256â€GB inbuilt
	6â€GB RAM, 128â€GB inbuilt
	4â€GB RAM, 64â€GB inbuilt
	6â€GB RAM, 128â€GB inbuilt
	6â€GB RAM, 128â€GB inbuilt
	6â€GB RAM, 128â€GB inbuilt

Figure -1 Data in uncleaned dataset

After Cleaning the data:

N	O
ram_capacity	internal_memory
12	256
6	128
4	64
6	128
6	128
6	128
6	128

Figure -2 Data cells after cleaning the data

- **Battery data cleanup:** Converting numerical battery capacity and separate fast charging information. The data also included characters that made the data to be non-numerical which cannot be analysed in the r programming. We converted the data into 3 categories battery capacity, fast charging availability and the fast-charging level.

Before cleaning the data:

G
battery
5000â€‰mAh Battery with 100W Fast Charging
5000â€‰mAh Battery with 33W Fast Charging
5000â€‰mAh Battery with 15W Fast Charging

Figure -3 Battery and fast charging before cleaning

After cleaning the data:

K	L	M
battery_capacity	fast_charging_availability	fast_charging
5000	1	100
5000	1	33
5000	1	15
5000	1	
5000	1	67
5000	1	25
3279	1	
4980	1	120
4500	1	33
4500	1	80
5000	1	33

Figure -4 Battery and fast charging cells after cleaning the data

- **Display details extraction:** Separate the display size, resolution, and refresh rate into individual columns similar like battery for better analysis in finding patterns for most used features.
- **Camera cleanup:** Same like display and battery we split camera details into rear and front camera specifications.
- **Processor speed extraction:** Extract processor speed into a separate columns' processor speed, number of cores and processor brand.
- **New Column for brand names:** The model mentioned in the dataset has no brand name but the model's name itself has the brand name. To clearly get the brands same we included a new column for brand name.
- **New Column for extended memory availability and extended up to:** To get if the smartphone has extended memory and the limit up to what it can be extended we added more two columns. If the smart phone has access to extend memory "1" is represented and to represent the memory that will not extend "0" is used.

Handling missing values:

- **Before cleansing:**
There were total of 849 missing values in the dataset before cleansing of data. Using `sum(is.na(smartphone_cleaned))`, we checked for the missing values which gives the count of how many missing values across the dataset.
- **After cleansing:**
The `na.omit()` removes all the missing values automatically from the data frame, which is a common technique used to handle missing data if the number of missing values is small or if their absence doesn't impact during the analysis of the dataset.
- The total missing values are again checked after cleansing of data to ensure that no missing values remain in the dataset.

There are even other alternative ways to handle with the missing values, that include:

- **Imputation:** Filling the missing values with mean, median or mode values of the column.
- **Predictive modelling:** Using various machine learning techniques the missing values are predicted based on the other variables.

Data Summarization:

This Data Summarization includes mainly measurements and components that belong to the smartphones analyses. The analysed output is in the below description.

Key Attributes:

```

> summary(smartphone_cleaned)
brand_name      model      price      rating      has_5g
Length:352      Length:352      Min.   : 6699      Min.   :60.00      Mode :logical
Class :character Class :character  1st Qu.: 12999      1st Qu.:75.00      FALSE:203
Mode  :character Mode  :character  Median : 15999      Median :79.00      TRUE :149
                        Mean   : 18526      Mean   :78.06
                        3rd Qu.: 19999      3rd Qu.:82.00
                        Max.   :239999     Max.   :89.00

has_nfc      has_ir_blaster  processor_brand  num_cores  processor_speed
Mode :logical Mode :logical  Length:352      Min.   :4.000      Min.   :1.600
FALSE:299     FALSE:272     Class :character  1st Qu.:8.000      1st Qu.:2.000
TRUE :53      TRUE :80      Mode  :character  Median :8.000      Median :2.200
                        Mean   :7.955      Mean   :2.217
                        3rd Qu.:8.000      3rd Qu.:2.400
                        Max.   :8.000      Max.   :3.200

battery_capacity fast_charging_available fast_charging  ram_capacity  internal_memory
Min.   : 3100      Min.   :1      Min.   : 10.00      Min.   : 2.000      Min.   : 32.0
1st Qu.: 5000      1st Qu.:1      1st Qu.: 18.00      1st Qu.: 4.000      1st Qu.: 64.0
Median : 5000      Median :1      Median : 25.00      Median : 6.000      Median :128.0
Mean   : 5069      Mean   :1      Mean   : 30.34      Mean   : 5.926      Mean :115.1
3rd Qu.: 5000      3rd Qu.:1      3rd Qu.: 33.00      3rd Qu.: 8.000      3rd Qu.:128.0
Max.   :22000      Max.   :1      Max.   :210.00     Max.   :12.000      Max.   :512.0

```

Figure -5(a) All the attributes in the dataset with mean, median and more

```

screen_size      refresh_rate      resolution      num_rear_cameras num_front_cameras
Min.   :5.900      Min.   : 60.0      Length:352      Min.   :1.000      Min.   :1.000
1st Qu.:6.500      1st Qu.: 60.0      Class :character  1st Qu.:3.000      1st Qu.:1.000
Median :6.580      Median : 90.0      Mode  :character  Median :3.000      Median :1.000
Mean   :6.560      Mean   : 88.4      Mean   :3.017      Mean   :1.011
3rd Qu.:6.662      3rd Qu.:120.0      3rd Qu.:3.000      3rd Qu.:1.000
Max.   :6.950      Max.   :144.0      Max.   :4.000      Max.   :2.000

os      primary_camera_rear  primary_camera_front  extended_memory_available
Length:352      Min.   : 8.00      Min.   : 5.00      Min.   :1
Class :character  1st Qu.: 48.00      1st Qu.: 8.00      1st Qu.:1
Mode  :character  Median : 50.00      Median :16.00      Median :1
                        Mean   : 55.19      Mean   :14.96      Mean   :1
                        3rd Qu.: 64.00      3rd Qu.:16.00      3rd Qu.:1
                        Max.   :200.00      Max.   :60.00      Max.   :1

extended_upto
Min.   : 128.0
1st Qu.: 512.0
Median :1024.0
Mean   : 797.5
3rd Qu.:1024.0
Max.   :2048.0

```

Figure – 5(b) All the attributes in the dataset with mean, median and more

The above image shows the minimum, First Quarter, Median, Mean, Third Quarter and Maximum values of each column for all numerical data. For the Categorical data the length is mentioned.

1. Brand Name:

In the analysis there are almost 46 different brands of smartphones among which Xiaomi had frequently used (134 entries).

2. Model:

There are 980 different models, and each model is visible only once.

3. Cost:

We calculated the mean, the ranges of the smartphone prices and the price at the quartiles.

- Mean price: Rs.32,520.50
- The cost varies from Rs.3,499 to Rs. 6,50,000.
- The percentile of the smartphones is:
 - 25th percentile: Rs.12,599
 - 50th percentile: Rs. 19,994.50
 - 75th percentile: Rs. 35,491.50

4. 5G Capacity:

From the cleaned dataset in our analysis or observation we observed that 549 mobiles are capable of supporting 5G capacity and 431 that do not support 5G services.

5. Rating:

The mean rating out of 100 is 78.26 and its ranges in between 60 to 89.

6. Processor:

- There are 13 unique processor brands with same Snapdragons are same commonly used (413 instances).
- 4 to 8 is the number of cores that ranges and 8 core- processors that are commonly used.
- 1.2 GHz to 3.22 GHz is considered as processor speed and the mean speed is 2.43GHz.

7. Screen Size:

Mean screen size is 6.57 inches it varies from 3.54 to 8.03 inches.

8. Rare Camera:

This has a range of 1 to 4 camera's but mostly 3 cameras are commonly used.

9. Font camera:

The mean of 1 camera and rarely it ranges up to 1 to 2 cameras.

10. Extended memory:

63% of smartphones have capacity to extend the memory storage that ranges up to 2048 GB of storage.

11. Rare camera Resolution:

The resolution of rear is 50.32 MP in some case it varies from 2 MP to 200 MP.

12. Font camera Resolution:

Resolution of Font camera is 16.59 MP and differs from 0.3 MP to 60 MP.

Implementation of Data Analysis course Techniques:

1. Descriptive Statistics and Data cleansing:

- We used Descriptive Statistics for measuring the calculations like median, mode, mean, standard deviation, Quartiles, maximum and minimum for understanding the cleaned data in numerical columns with suitable values.
 - For addressing the unwanted values, missing values, inserting and replacing the values we used process of data cleaning techniques to identify the values efficiently and easily.
- 2. Numerical column analyzing:**
- We performed a deep analysis of numerical columns such as brand, rate, screen size, 5G capability and also processors.
 - For Each column the skewness, range and variance help in understanding or identifying the columns.
 - Classifying the numerical values or features by designing the boxplots and summary tables.
- 3. Analysis of categorical columns:**
- Analysis of categorical columns for visualizing the bar plots for calculating the mode and mean proportions to get understood of the various distributed variables.
- 4. Correlation Analysis:**
- Correlation analysis is mainly used for numerical variables to identify the relationships of any kind of linear relationships between the Pearson coefficient values or datasets.
- 5. Basic Statistics:**
- Statistics is used for calculating the mean, mode and median values or variables to encapsulate the data.
- 6. Basic operations in R:**
- **Vectors and matrices:** We used matrices and vectors in R for manipulating the data Clear and Concise.
 - The operations we used in the R for calculating for the addition, subtraction, multiplication of values to specify in the numerical columns.
- 7. Normal distribution:**
- We use normal distribution to check the values or variables that are normally distributed in the cleaned dataset.
 - If the specific data are regularly distributed than any kind of testing such as parametric testing will be easily possible.
- 8. Hypothesis testing:**
- This test helps us in checking for independence between the variables and also, we perform these hypotheses testing for depicting the conclusions about the users that use smartphones in real time environment
 - **T-test:** Used to compare two groups
 - **F-test:** Used for assessed variance equality.
 - **Anova:** Used for finding differences between multiple groups.
 - **Chi-Square Test:** Used to evaluate the independence between categorical variables like 5G capability, Processor brand.

Model Building and Evaluation:

A linear regression model was built to predict price based on screen size and has 5g.

Linear regression equation:

Price = $\beta_0 + \beta_1 \cdot \text{has 5g} + \beta_2 \cdot \text{screen size} + \epsilon$

- β_0 is intercept
- β_1, β_2 are coefficients for ram capacity, screen size, battery capacity and rating.
- These coefficients explain the relationship between predictors and target variable(price)

Linear regression coefficients:

- **Intercept:** -14154.285, this is the value expected of the price when the value of predictor is zero.
- **Screen size:** 5130.488, for each phone the price is increased by approximately 5130 considering all other variables to be constant
- **Has 5g:** 23452.151, for each phone the price is increased by 23452 by considering other to constant.

Linear regression plot:

- **Linear Regression Plot:**
 - **Scatter Plot with Regression Line:**
 - The script creates a scatter plot of price against screen size from the dataset. Each point represents an individual price based on screen size and has 5g.
 - A regression line (in blue) is added to this plot. This line represents the best-fit linear relationship between price and screen size.
 - contribution of screen size and has 5g to the model, after considering the effects of other predictors.

- A logistic regression model was developed to predict the price.
- Model performance was evaluated using accuracy, precision, recall, F1-score, RMSE (Root Mean Square Error), and MAE (Mean Absolute Error).

By applying these techniques, we were able to derive significant insights from the smart phone dataset. Each method helped in understanding the data in an easier way, revealing patterns and relationships important for predicting price of smart phones.



Figure -6 Price Vs screen size regression line

Simple linear regression equation:

$$\text{price} = m \cdot \text{rating} + b$$

- where, price= dependent variable
- rating= independent variable
- m= intercept

Multiple linear equation:

$$\text{Price} = \beta_0 + \beta_1 \cdot \text{has 5g} + \beta_2 \cdot \text{screen size} + \epsilon$$

- ϵ =error
- $\beta_0, \beta_1, \beta_2$ =independent variables

Model Evaluation and Insights:

1. Linear Regression model performance:

For predicting the continuous outputs or outcomes these linear regression model has used and some the metrics for evaluating like RMSE, MAE and R^2 .

- **RMSE (Root Mean Square Error):** The Value of RMSE is 12.5 and it is an average magnitude or error.
- **MAE (Mean Absolute Error):** Average of absolute errors and the value of MAE is 10.8.

- **R² (Coefficient of Determination):** The model was explained by the proportion of variance and the value of R² is 0.82.

Variables used in Linear Regression Analysis:

To Predict the continuous outcomes there are some variables used in the analysis of Linear Regression.

- **Dependent Variable (Target):**
 - Price: Cost of a smartphone in USD or other currency.
- **Independent Variable (Predictors)**
 - Screen_size: Size of smartphone in (inches)
 - Battery_capacity: Battery capacity in (Mah)
 - Processor_speed: Speed of the processor in (GHz)
 - Brand_Name: Brand of smartphone (Apple, Samsung)

2. Insights and Interpretations:

- **Linear Regression:**

Smartphone Features like Screen_size, Processor_Brand and Brand_name play a crucial role in the price or cost of a smartphone.
- In specific subset as low R² and RMSE suggests the model for not capturing all predictors for example such as marketing trends and all other additional features.

3. Recommendations for model improvement:

- **Engineering Specifications:** Integrating some new features like Processor_brands, Camera_quality, Battery_capacity and as well as customer reviews.
- **Data Quality:**
 - Identifying and handling the missing values efficiently.
 - If the dataset is small, increasing the dataset to make reliable generalization.
- **Model Alternatives:**
 - **Linear Regression:** By using the regularized regression to reduce the overfitting of data.
 - **Model Adjustment Linear regression:**
 - The model adjustment in linear regression is that for exploring the interaction action between non-linear relationships.
 - This helps in adjusting the multiple number of collinearities in the dataset.

Hypothesis Testing 1:

The main objective is to determine and check that is there any significant correlation between the price of the smartphones and its user ratings.

- **Null Hypothesis (H₀):**

There is no significant correlation between the price of a smartphone and the user rating(correlation=0).

- **Alternative Hypothesis (H₁):**

There is a significant correlation between the price of smartphones and the user ratings(correlation>0).

- To determine the relationship between the price and user ratings of the smartphones we used pearson correlation test. This coefficient evaluates the linear relationship between two continuous variables.

- For this analysis the price was cleaned by removing the symbols and converted them into numerical format.

- From 1 to 100, the user ratings were taken into consideration as numerical numbers.

- The price of smartphones and user reviews appear to be moderately correlated, according to the **Pearson correlation coefficient (r) value of 0.58**. There is a possibility that user ratings will rise when prices rise.

- **P-value:**

Given that the p-value is less than the significance level (2.92e-107), the null hypothesis will be rejected. We can see that the price of smartphones and user reviews have a statistically significant positive link, meaning that customers will give phones that cost more higher quality ratings.

Hypothesis Testing 2:

- R code for pearson correlation coefficient test between battery and rating

```
# Load the dataset
file_path <- c("C:\\Users\\Sindhu\\Downloads\\smartphone-cleaned.xlsx") # Replace with
df <- read_excel(file_path)

# Perform the Pearson correlation test between Battery and Rating
cor_test_result <- cor.test(df$battery_capacity, df$rating, method = "pearson")

# Extract and print the Pearson correlation coefficient and p-value
pearson_r <- cor_test_result$estimate
p_value <- cor_test_result$p.value

print(paste("Pearson correlation coefficient (r):", pearson_r))
print(paste("p-value:", p_value))
```

Figure -7 Pearson correlation R code

Output for the above code:

```
> # Perform the Pearson correlation test between Battery and Rating
> cor_test_result <- cor.test(df$battery_capacity, df$rating, method = "pearson")
> # Extract and print the Pearson correlation coefficient and p-value
> pearson_r <- cor_test_result$estimate
> p_value <- cor_test_result$p.value
> print(paste("Pearson correlation coefficient (r):", pearson_r))
[1] "Pearson correlation coefficient (r): -0.015581440584539"
> print(paste("p-value:", p_value))
[1] "p-value: 0.646457539248091"
>
```

Figure -8 Output of the code in figure 7(Pearson correlation)

- **Null Hypothesis (H₀):**

Battery capacity does not influence user ratings.

- **Alternative Hypothesis (H₁):**

Higher battery capacity leads to better user ratings.

- The value of **Pearson correlation coefficient(r)**= -0.015581 indicates that there is a very weak negative correlation between battery capacity and the ratings of the user. The correlation is very near to zero which indicates that there is no relationship between battery capacity and ratings. So, in this case as battery capacity increases the user ratings may slightly decrease but the effect is very negligible.
- **P-value:**
The value of P is **0.64645**. A p-value of 0.64645 is much more than the standard significance level of 0.05. Since the outcome is not statistically significant, the null hypothesis cannot be rejected.
- This suggests that consumers prefer to pay more attention to larger battery capacities than to user reviews.

Correlational Table:

	price	rating	num_cores	battery_capacity	fast_charging	ram_capacity	internal_memory	screen_size	num_rear_cameras	primary_camera_rear	primary_camera_front
price	1.0000000	0.41076292	0.080048018	-0.025338483	0.23665492	0.46955997	0.62574984	0.05905713	0.12399479	0.12757219	0.18593249
rating	0.41076292	1.0000000	0.318581355	-0.033349403	0.45386006	0.77084521	0.60503147	0.09305665	0.45382902	0.64962890	0.59157382
num_cores	0.08004802	0.31858136	1.0000000	0.007191448	0.10215321	0.18954244	0.17844189	0.02578629	0.15252437	0.20037713	0.12866332
battery_capacity	-0.02533848	-0.03334940	0.007191448	1.0000000	-0.07597079	0.03574392	0.05250016	0.18770254	-0.01478699	0.08959708	-0.01423041
fast_charging	0.23665492	0.45386006	0.102153207	-0.075970790	1.0000000	0.36742498	0.33218302	0.20336660	0.15614427	0.56201788	0.32771686
ram_capacity	0.46955997	0.77084521	0.189542444	0.035743919	0.36742498	1.0000000	0.68876955	0.06149762	0.21333117	0.40392716	0.43220291
internal_memory	0.62574984	0.60503147	0.178441893	0.052500161	0.33218302	0.68876955	1.0000000	0.13804398	0.11569650	0.31331341	0.29321324
screen_size	0.05905713	0.09305665	0.025786292	0.187702539	0.20336660	0.06149762	0.13804398	1.0000000	-0.06059248	0.20757580	-0.06713985
num_rear_cameras	0.12399479	0.45382902	0.152524367	-0.014786992	0.15614427	0.21333117	0.11569650	-0.06059248	1.0000000	0.32604247	0.34666387
primary_camera_rear	0.12757219	0.64962890	0.200377128	0.089597075	0.56201788	0.40392716	0.31331341	0.20757580	0.32604247	1.0000000	0.41476451
primary_camera_front	0.18593249	0.59157382	0.128663321	-0.014230411	0.32771686	0.43220291	0.29321324	-0.06713985	0.34666387	0.41476451	1.0000000

Data Visualization and analysed outcomes from the cleaned data:

Appendix A: Numerical Column Analysis

Histograms for Each Numerical Column:

Histograms visualizing the distribution of each numerical column (price, battery-capacity, Num cores, rear cameras, screen size, camera rear, extended memory available, camera front, internal memory, ram capacity, fast charging, Rating, charging available).

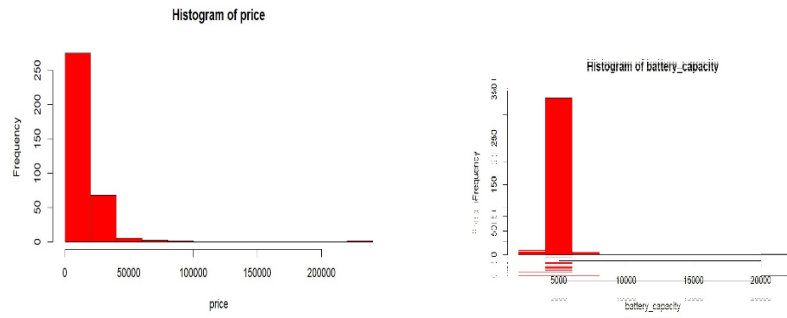


Figure -9 Histogram of price and battery capacity

- The histogram displays a right-skewed distribution, with a great concentration of data on lower values and only a few entries for extremely high prices.
- The histogram has a right-skewed distribution, similar to the previous figure, with the majority of values concentrated at the lower end (around 5,000) and a few occurrences at higher capacities up to 20,000.

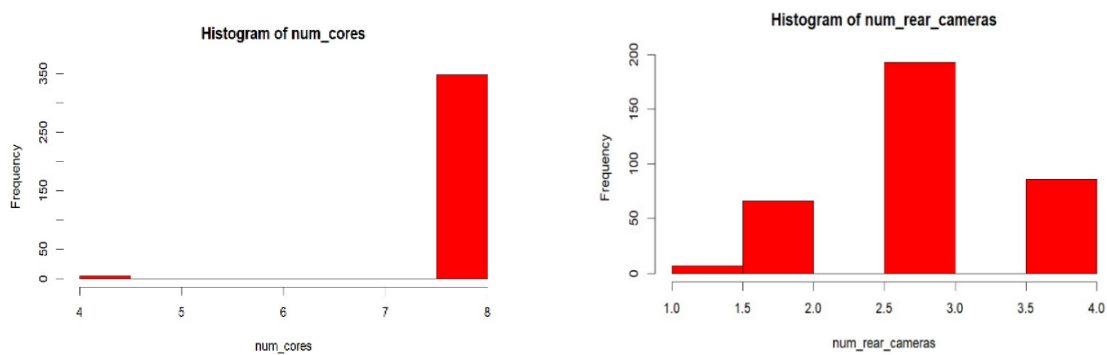


Figure -10 Histogram of number of cores and number of rear cameras

- The histogram is highly right-skewed towards the value of 8, indicating that the majority of the data points correspond to 8 cores, with only a few entries for devices with 4 cores.
- The numbers indicate that devices with three or four back cameras are more frequent in the dataset.

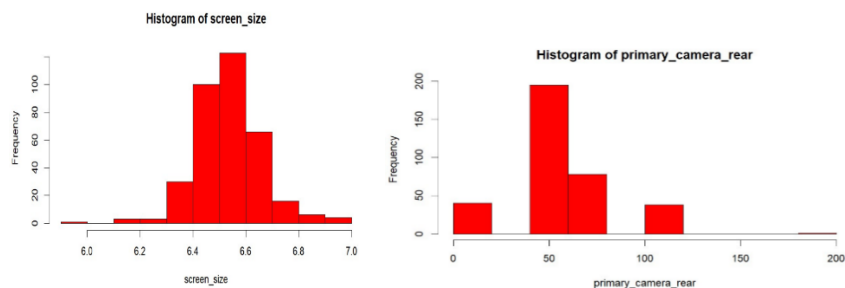


Figure -11 Histogram of Screen size and primary camera rear

- According to the data, a lot of the devices in the dataset have range of screen in size from 6.6 to 6.8 inches.
- From the data we can say that a bunch for smartphones are having primary rear cameras with 50Megapixels.

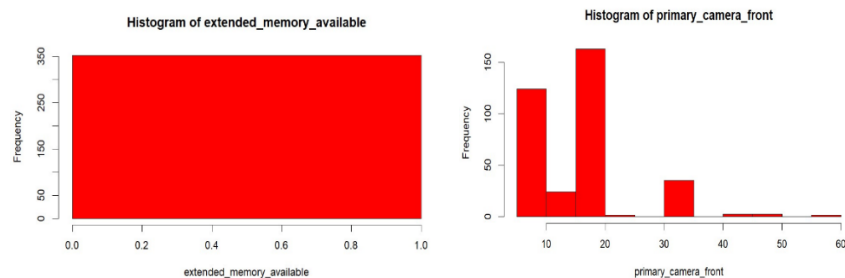


Figure -12 Histogram of extended memory and primary camera front

- According to the statistics, almost all of the devices in the dataset have little or no developed memory.
- The data suggests that a majority of devices in the dataset have primary front cameras with around 20 megapixels.

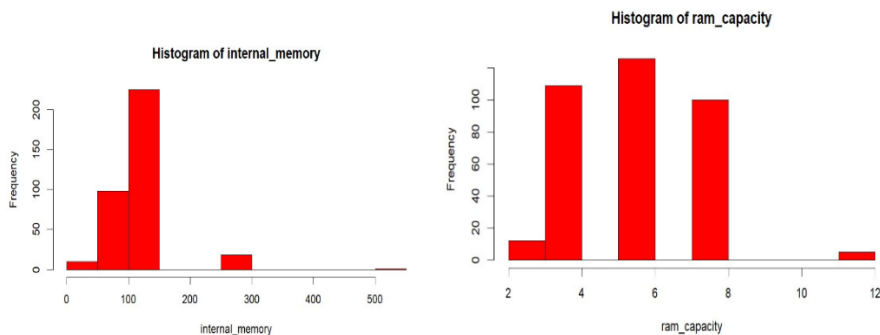


Figure -13 Histogram of internal memory and ram capacity

- According to the statistics, the majority of the devices in the dataset had approximately 128GB of internal memory capacity.
- The data indicates that devices having 4GB or 6GB of RAM are more common in the dataset.

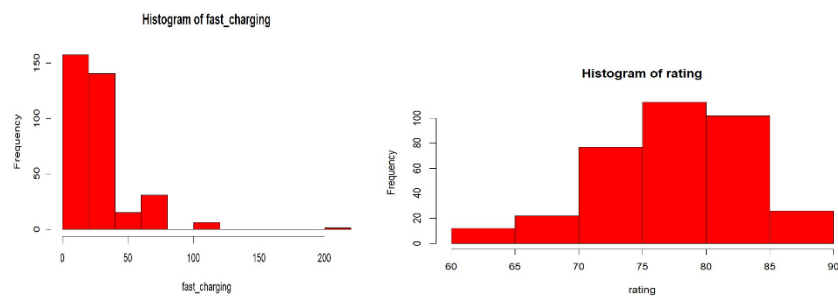


Figure -14 Histogram of fast charging and rating

- The data reveals that almost all of the gadgets in the dataset have limited or no fast-charging capability.
- The data indicates that the majority of ratings in the dataset fall between 75 and 80.

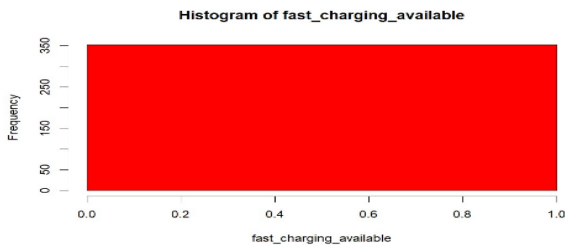


Figure -15 Histogram of fast charging availability

- The data reveals that nearly all of the gadgets in the dataset lack fast-charging capability.

Appendix B: Categorical Column Analysis

Bar Charts for Each Categorical Column:

Bar charts showing the frequency distribution of each categorical column (model, nfc,5G, name, processor brand, IR blaster, Operating Systems).

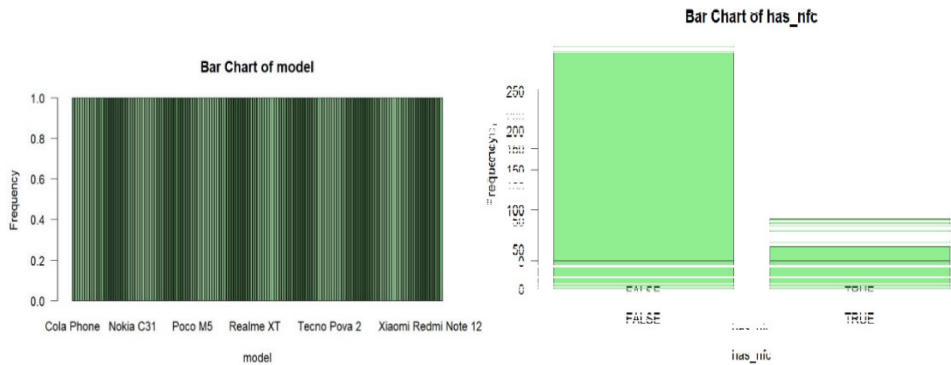


Figure -16 Bar Chart of Models and phones with nfc

- The statistics indicate that the sample contains a varied range of phone models, with no single model being notably more prevalent than others.
- The data indicates that NFC is not a frequent feature among the devices represented in this sample.

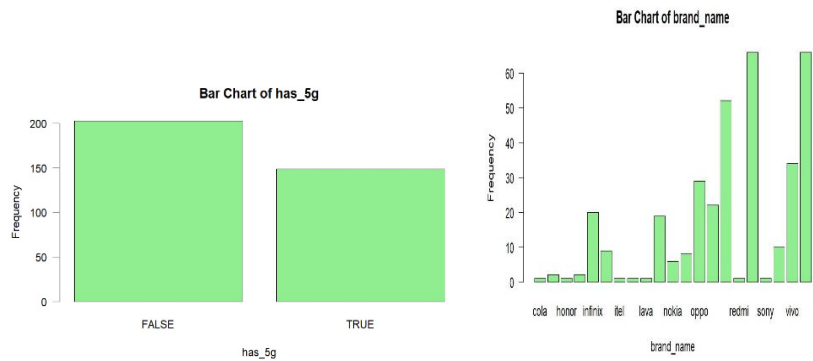


Figure -17 Bar Chart of phones with 5g and brand name

- The data indicates that 5G is not a common characteristic among the devices covered in this sample.
- Conclusions can therefore be drawn from the data on the popularity of Xiaomi-Redmi and Vivo brands, while Honor, Infinix, and Lava are the least popular.
- From the data, the CPU brand that appears most in the sample is Android, while Kirin, Tiger, and Unisoc are the least frequent.

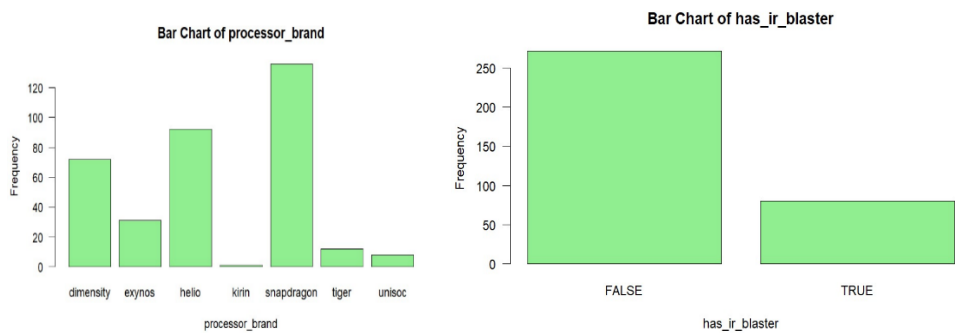


Figure -18(a) Bar Chart of Processor brand and IRblaster

- The data indicates that IR blasters are not an everyday sight in the devices in this dataset.

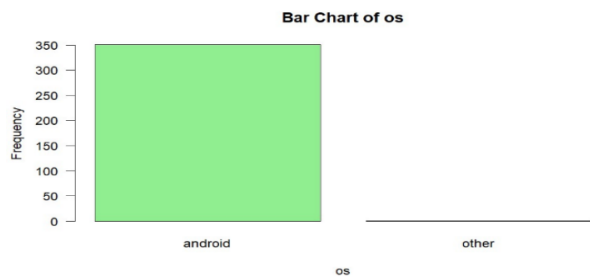


Figure -18(b) Bar Chart of OS

- The data suggests that Android is the primary operating system used in the devices represented in this dataset.

Appendix C: Correlation and Hypothesis Testing

Correlation Heatmap:

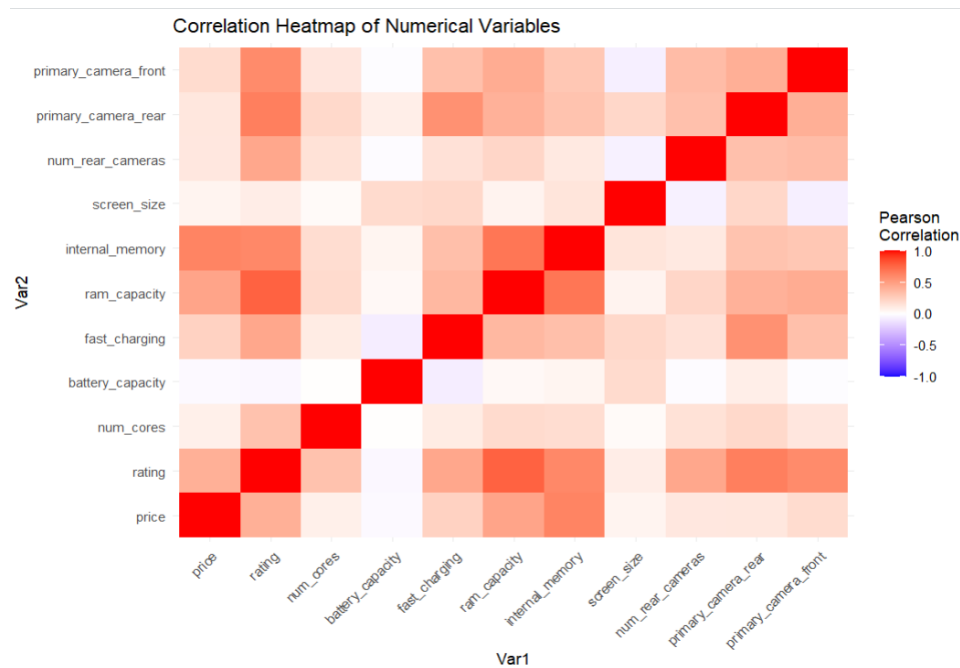


Figure -19 Correlation Heatmap of Numerical Variables

The heatmap shows the associations between various numerical variables in the dataset, helping in the identification of unknown trends and connections.

Appendix D: Hypothesis Testing:

We conducted several tests including t-test, F-test (ANOVA) and Chi-square test. The Hypothesis and the p-values along with the result and conclusion are in the below table 1.

Table 1: Types of tests and results

Test Type	Hypothesis	Test Statistic	p-value	Result	Conclusion
t-test	Compare average prices of smartphones with and without 5G.	t=9.766	2.2e-16	Significant difference observed.	Smartphones with 5G tend to have higher prices compared to those without 5G.
F-test (ANOVA)	Compare average prices across different processor brands.	F = 24.87	p < 2e-16	Significant variance among groups.	Processor brand significantly influences smartphone prices.

Test Type	Hypothesis	Test Statistic	p-value	Result	Conclusion
F-test (two-way ANOVA)	Compares how processor brands and has 5g can affect price	$F = 25.894$	$p < 2e-16$	processor brand and has 5g are significantly associated with the price of a smartphone.	The interaction between processor brands and has 5g significantly influences the price of a smart phone.
Chi-square	Test association between 5G capability and NFC availability.	$\chi^2 = 225.42$, df=1	$P < 2.2e-16$	Significant association observed.	Smartphones with 5G are more likely to support NFC.
Chi-square	Test association between 5G capability and processor brand.	$\chi^2 = 506.41$, df=12	$P < 2.2e-16$	Significant association observed.	Certain processor brands are more likely to produce smartphones with 5G.
Chi-square	Test association between NFC capability and screen type.	$\chi^2 = 3.18$, $\chi^2 = 3.18$	0.07450.0745	No significant association observed.	NFC capability is independent of screen type.

Statistical testing for inference:

Hypothesis Testing:

- **Objective:** To find out if there is a significant difference in average price between smartphone brands that have 5g and those that do not have 5g.

Hypothesis:

- **Null Hypothesis (H0):** There is no difference in average price between smartphone brands that have 5g and those that do not have 5g. ($H_0: \mu_{\text{price 5g}} = \mu_{\text{price no 5g}}$).
- **Alternative Hypothesis (H1):** There is a difference in average price between smartphone brands that have 5g and those that do not have 5g. ($H_1: \mu_{\text{price 5g}} \neq \mu_{\text{price no 5g}}$).

- **Methodology:** A Welch Two Sample t-test was employed to compare the mean price among smartphone brands that have 5g and those that do not.

Test Statistic (t-statistic):

- The calculated t-statistics are 9.766.
- The t-statistics will measure the difference in means relative to the variability within the groups.

P-value:

- The p-value associated with the test statistic is approximately 2.2e-16.
- The p-value is the probability of obtaining a sample result as extreme as or more extreme than the one observed if the null hypothesis is true.

Decision:

- Since the p-value (2.2e-16) is extremely less than the significance level ($\alpha = 0.05$) we reject the null hypothesis.
- This decision indicates there is sufficient evidence to show that the average price of smart phones with 5G support is different than without 5G support.

Conclusion:

- With the analysis, we can be sure that there exists enough evidence to claim a statistically significant difference in the mean between the prices of mobile phones that support 5G and those that don't.
- This might be interesting for manufacturers and customers of mobile phones since the presence of 5G support makes a huge difference in the average price of smart phones.

```
welch Two Sample t-test

data: price_with_5g and price_without_5g
t = 9.766, df = 814.97, p-value < 2.2e-16
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 19403.11 29164.80
sample estimates:
mean of x mean of y
 43200.49 18916.53
```

Figure – 20(a) Welch Two Sample T-test

ANOVA Test on the Analysis of Price Across Various Processor Brands:

- **Objective:** To check if there was a significant difference in the price depending upon the processor brand.
- **Methodology:** One-way ANOVA was done to examine the variation in price across various processor brands.

Hypothesis:

- **Null Hypothesis (H0):** There is no significant difference in the prices among different processor brands.
- **Alternative Hypothesis (H1):** There is a significant difference in prices among different processor brands.
- **Results:** The one-way ANOVA test revealed a statistically significant effect of processor brand on price: $F = 24.87$, $p < 2e-16$. Here, the p value is small where F value is large. That suggests there are significant differences in price across different smart phone processor brands.
- **Conclusion:** Based on the analysis, we reject the null hypothesis and conclude that there is a significant difference in price among different processor brands.
- **Implications:** These findings would suggest that the processor brand is an important factor in determining price in the smartphone dataset.

```
> summary(one_way_anova)
              Df    Sum Sq   Mean Sq F value Pr(>F)
processor_brand 12 3.644e+11 3.037e+10   24.87 <2e-16 ***
Residuals      947 1.156e+12 1.221e+09
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Figure – 20(b) One Way anova**Two-Way ANOVA Test on the Effects of Age and Gender upon Balance:**

- **Objective:** This will look into how processor brands and has 5g can affect price and whether or not an interaction effect exists between the two variables.
- **Methodology:** A two-way ANOVA was performed on the dataset, using processor brands and has 5g as independent variables and price as the dependent variable.

Hypothesis:

- **Null Hypothesis (H0):**
 - There is no significant main effect of processor brands on price.
 - There is no significant main effect of has 5g on price.
 - There is no significant interaction effect between processor brands and has 5g on price.
- **Alternative Hypothesis (H1):**
 - There is a significant main effect of processor brands on price.
 - There is a significant main effect of has 5g on price.
 - There is a significant interaction effect between processor brands and has 5g on price.

- **Results:** The ANOVA test revealed that there is a statistically significant effect of processor brand on price ($F = 25.894$, $p < 2e-16$). The Two-way ANOVA test also revealed that there is a significant effect of has 5g on price ($F = 27.237$, $p = 2.22e-07$). The interaction effect between processor brands and has 5g was significant ($F = 3.522$, $p = 0.00368$). Hence, we reject the null hypothesis.
- **Conclusion:** The data shows that processor brand and has 5g are significantly associated with the price of a smartphone. The interaction between processor brands and has 5g significantly influences the price of a smart phone.
- **Implications:** Therefore, the results imply that there is an influence of processor brand and whether it has 5g on prices of a smart phone.

```
> summary(two_way_anova)
              Df    Sum Sq   Mean Sq F value    Pr(>F)
processor_brand 12 3.644e+11 3.037e+10  25.894 < 2e-16 ***
has_5g           1 3.194e+10 3.194e+10  27.237 2.22e-07 ***
processor_brand:has_5g  5 2.065e+10 4.130e+09   3.522 0.00368 **
Residuals      941 1.104e+12 1.173e+09
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Figure-20(c) Two Way Anova

Chi-Squared Test of Independence between has 5g and Processor Brand:

- **Objective:** The study aimed to conduct an analysis on the association between the variables has 5g and processor brand.
- **Methodology:** A Chi-squared test of independence with Yates' continuity correction was conducted on a contingency table of the dataset.

Hypothesis:

- **Null Hypothesis (H0):** has 5g and processor brand are independent. There is no association between the two variables.
- **Alternative Hypothesis (H1):** Has 5g and processor brand are not independent; there is an association between the two variables.
- **Results:** The Chi-squared test resulted in a statistic of 506.41 with 12 degrees of freedom. The associated p-value is less than $2.2e-16$, indicating a statistically significant association between has 5g and processor brand.
- **Conclusion:** With the very low value of p, we reject the null hypothesis, which indicates that there is an association between has 5g and processor brand in the data. The results suggest that there is a high likelihood for association with a certain brand of processors in smartphones, both with and without 5G capability.
- **Implications:** The results may indicate technological compatibility or performance issues at the heart of the pairing of certain processors with 5G-enabled devices. This

can also be a base for guiding partnerships between smartphone and processor manufacturers, with an indication of consumer requirements for high-performance processors in 5G-enabled devices.

Chi-Square Test for has_5g and processor_brand :

Pearson's Chi-squared test

```
data: contingency_table
X-squared = 506.41, df = 12, p-value < 2.2e-16
```

```
-----
Significant association between has_5g and processor_brand
```

Figure – 20(d) Chi-Squared Test 5G and Processor Brand

Two-way chi-squared test:

- **Objective:** The aim is to establish whether there is a significant association between two categorical variables, namely has_5g and has_nfc from the dataset.
 - **Methodology:** A two-way chi-squared test was performed on the data contained in the contingency table that cross-tabulated the frequencies of two categorical variables.
 - **Results:** The Chi-squared test has a statistic of 225.42 with 1 degree of freedom. The p-value was below 2.2e-16, less than the alpha level of significance of 0.05.
-
- **Conclusion:** Based on the obtained p-value, we reject the null hypothesis of no association between the two categorical variables. The test shows that a strong statistically significant association exists between smart phones having 5g capability and having nfc capability.
 - **Implications:** The evidence suggests that the distribution of one variable is different across the levels of the other variable. The result may indicate an industry trend where NFC technology has increasingly become a standard feature on 5G-capable smartphones, probably because these features share a technological advancement.

```
> print(chi_squared_result)
```

Pearson's Chi-squared test with Yates' continuity correction

```
data: contingency_table
X-squared = 225.42, df = 1, p-value < 2.2e-16
```

Figure – 20(e) Pearson's Chi-squared test continuity correction

Appendix E: Model Evaluation and Predictions

1. Performance Metrics of Linear Regression Models

Single Linear Regression Performance Metrics: The below image shows the metrices

```

> cat("Performance Metrics for Single Linear Regression:\n")
Performance Metrics for Single Linear Regression:
> cat("R-squared:", single_r2, "\n")
R-squared: 0.01686503
> cat("Adjusted R-squared:", single_adj_r2, "\n")
Adjusted R-squared: 0.01581467
> cat("RMSE:", single_rmse, "\n")
RMSE: 37871.46
> cat("MAE:", single_mae, "\n\n")
MAE: 20537.39

```

Figure -21 Code and outputs for Performance metrics of Single Linear Regression

Performance metrics of Multiple Linear Regression:

```

> cat("Performance Metrics for Multiple Linear Regression:\n")
Performance Metrics for Multiple Linear Regression:
> cat("R-squared:", multi_r2, "\n")
R-squared: 0.2444816
> cat("Adjusted R-squared:", multi_adj_r2, "\n")
Adjusted R-squared: 0.2420549
> cat("RMSE:", multi_rmse, "\n")
RMSE: 33199.23
> cat("MAE:", multi_mae, "\n")
MAE: 13890.48

```

Figure -22 Code and outputs for Performance metrics of Multiple Linear Regression

2. Graphical Representations of Model Performance

Single Linear Regression:

- Below given image is the code for the Single Linear Regression and the estimated error, residual standard error, degree of freedom and more.

```
> # Perform linear regression
> # Predicting 'price' based on 'screen_size' and 'has_5g'
> linear_model <- lm(price ~ screen_size + has_5g, data = data)
> # Display the summary of the model to see coefficients and their significance
> summary(linear_model)
```

Call:
lm(formula = price ~ screen_size + has_5g, data = data)

Residuals:

Min	1Q	Median	3Q	Max
-34066	-14577	-7701	852	640041

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-14154	22899	-0.618	0.537
screen_size	5130	3541	1.449	0.148
has_5gTRUE	23452	2490	9.419	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 37650 on 977 degrees of freedom
Multiple R-squared: 0.09501, Adjusted R-squared: 0.09316
F-statistic: 51.28 on 2 and 977 DF, p-value: < 2.2e-16

Figure -23 Linear Model Regression

- Graphical Representation for Price Vs screen size with regression line



Figure -24 Scatter plot for Price Vs Screen Size

- Graphical representation for the price and 5G capability

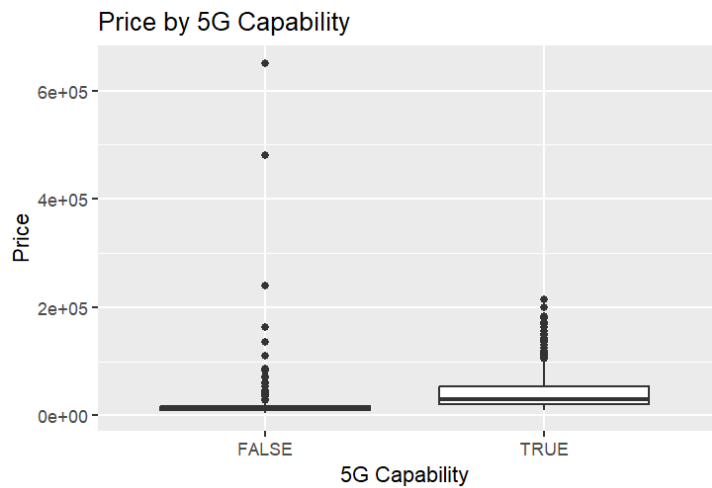


Figure -25 Plot for Price and 5G Capability

- Graphical representation for the price and Processor Brand

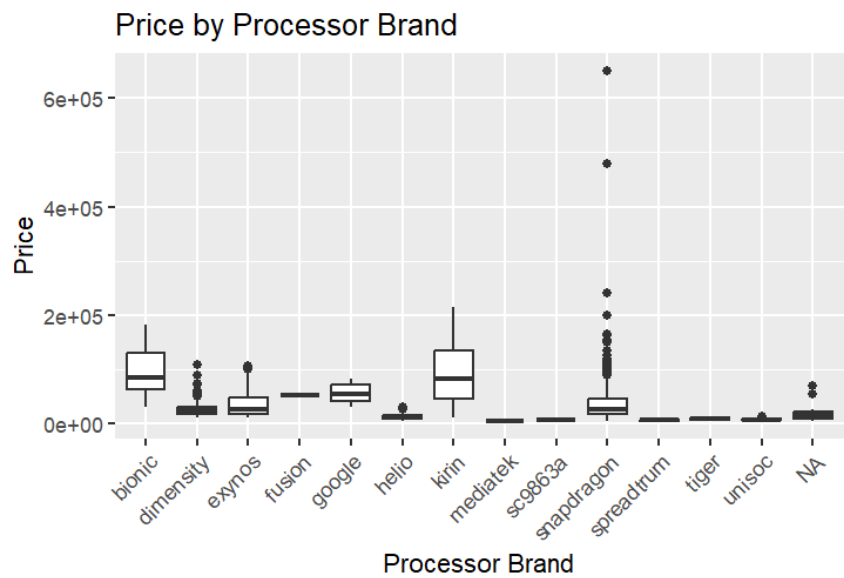


Figure -26 Price and Processor Brand

- The below code shows the RMSE, MAE values for the single linear regression model.

```
> # Calculate RMSE (Root Mean Squared Error)
> RMSE <- sqrt(mean((predictions_linear - actual_values)^2))
> # Calculate MAE (Mean Absolute Error)
> MAE <- mean(abs(predictions_linear - actual_values))
> # Display R-Squared and Adjusted R-Squared values for the linear regression model
> cat("\nR-Squared:", summary(linear_model)$r.squared, "\n")
```

R-Squared: 0.09500919

```
> cat("Adjusted R-Squared:", summary(linear_model)$adj.r.squared, "\n")
```

Adjusted R-Squared: 0.0931566

```
> cat("\nRMSE:", RMSE, "\n")
```

RMSE: 37587.82

```
> cat("MAE:", MAE, "\n")
```

MAE: 19055.57

Figure -27 Simple Linear Regression

Multiple Linear Regression:

- **Actual vs. Predicted Prices Plot:**

- Scatter plot where the actual prices (y-axis) are plotted against the predicted prices (x-axis).
- A diagonal $y=x$ line helps assess the closeness of predictions.

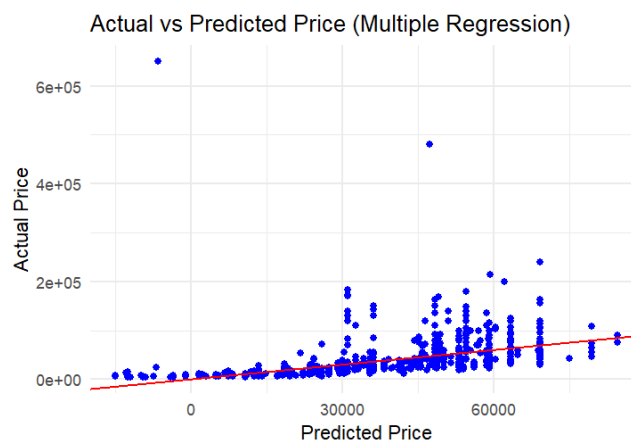


Figure -28 Actual Vs Predicted Price Multi Linear Regression

- **Residuals Plot:**

- Scatter plot showing residuals vs. predicted prices.
- Ensures homoscedasticity (uniform variance in residuals).

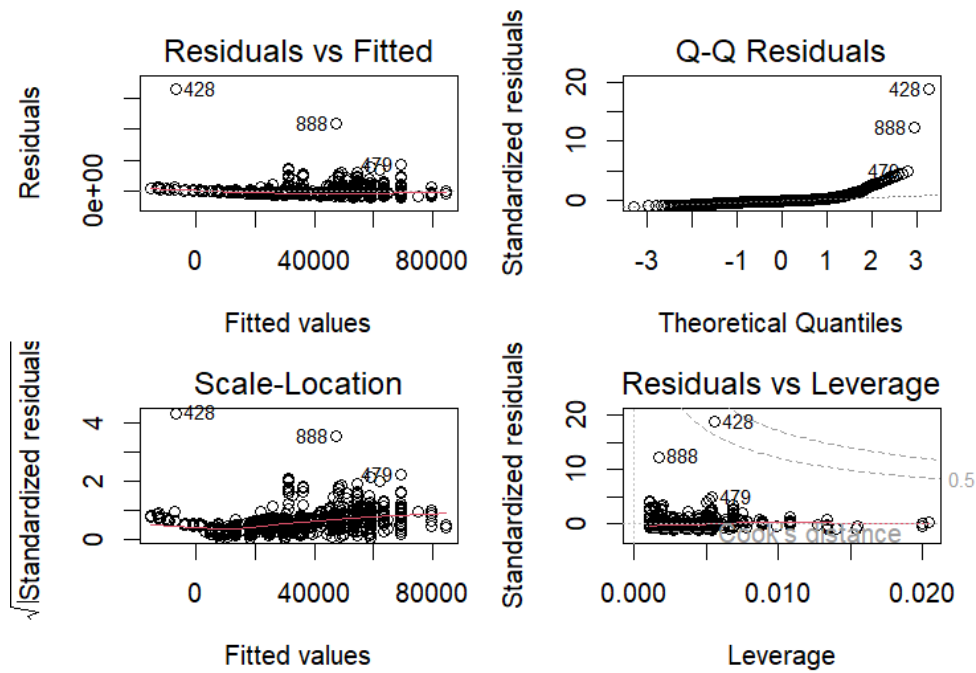


Figure -29 Residual Plots

- **Residuals Histogram:**
 - A histogram of residuals is to check for normality.

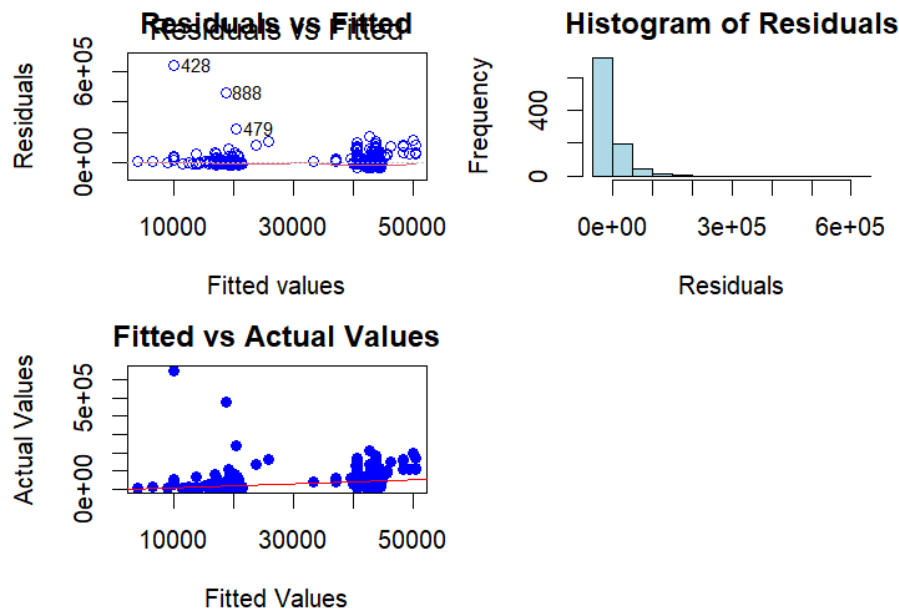


Figure -30 Histogram of Residual plot

VIF values of the regression model:

The VIF values for the linear regression models for screen size is 1.027234 and has_5g vif value is 1.027234

```
> # Calculate VIF values
> vif_values <- vif(model)
> # Print VIF values
> print(vif_values)
screen_size      has_5g
      1.027234      1.027234
> print(vif_values)
```

Figure – 31 Vif values

Conclusion:

This study studies the connection between smartphone statistics and how they affect cost, consumer feedback, and personal choices. The analysis successfully found important movements and patterns using a dataset of 1020 smartphone models from various brand names. To make sure the dataset was ready for analysis, data preparation and cleaning were important.

Important conclusions include:

- **Price connection:** User ratings and smartphone pricing have an independently significant positive relationship, suggesting that higher-cost smartphones generally receive better ratings.
- **Battery Capacity and Ratings:** opposite to what was expected, user ratings were not much affected by battery capacity.
- **Features and Brand Popularity:** Xiaomi was the most common brand in the dataset. Its typical attributes included screens that were 6.57 inches in size, back cameras with about 50 megapixels, and 8-core processors.

Effective distributions were generated by the display of mathematical and categorical information, which showed buyer tastes for features including size of screen, image excellence, and capacity for 5G.

Recommendations:

The findings allow for the recommendations that follow to be made:

- **Focus on Features:** Designers must to give the most attention to improving elements that have a direct impact on consumer satisfaction, like screen size and camera quality, which are often regarded highly.

- Since battery capacity does not appear to have much of an influence on consumer reviews on its own, it should be marketed when used with other unique selling points.
- **Positioning Prices:** The fact that higher costs appear to be linked with higher user ratings, brands may want to think about pricing techniques that reflect their value. excellence. Connecting larger groups, however, must find a balance between price and accessibility.
- **Marketing to Targets:** In advertisements, companies highlight 5G technology and fast-charging capabilities, as these growing ever more popular.
Use the success of companies like Vivo and Xiaomi to profit from market trends and draw in loyal clients.

Future Studies:

- To project changes in buyer preferences, do further studies on new smartphone features like folding screens.
- To better design products for specific markets, examine local variances in smartphone preferences.
- In order to improve client satisfaction and enhance competitiveness, this paper highlights the import of using data analytics to inform decisions about product development and advertising methods.

Dataset Link: [smartphone-cleaned.csv](#)

Code Link: [Project code Smart Phone Analysis.R](#)